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import numpy as np
import matplotlib.pyplot as plt

# Input range
x = np.linspace(-10, 10, 400)

# Activation functions
def sigmoid(x):
    return 1 / (1 + np.exp(-x))

def tanh(x):
    return np.tanh(x)

def relu(x):
    return np.maximum(0, x)

def leaky_relu(x, alpha=0.1):
    return np.where(x > 0, x, alpha * x)

def softmax(x):
    exp_x = np.exp(x - np.max(x))    # ✅ stability fix
    return exp_x / np.sum(exp_x)

# Create subplots
plt.figure(figsize=(12, 8))

# Sigmoid
plt.subplot(3, 2, 1)
plt.plot(x, sigmoid(x))
plt.title("Sigmoid Activation Function")
plt.grid(True)

# Tanh
plt.subplot(3, 2, 2)
plt.plot(x, tanh(x))
plt.title("Tanh Activation Function")
plt.grid(True)

# ReLU
plt.subplot(3, 2, 3)
plt.plot(x, relu(x))
plt.title("ReLU Activation Function")
plt.grid(True)

# Leaky ReLU
plt.subplot(3, 2, 4)
plt.plot(x, leaky_relu(x))
plt.title("Leaky ReLU Activation Function")
plt.grid(True)

# Softmax (illustrative)
plt.subplot(3, 2, 5)
plt.plot(x, softmax(x))
plt.title("Softmax Activation Function (1D)")
plt.grid(True)

# Title (FIXED)

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plt.suptitle(" Activation Functions in Neural Networks ", fontsize=14)

plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.show()
```